

Simulation Parameterization Module (SPHM)

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Independent Methodological Authority

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: SIMULOS® — Simulation Governance Architecture

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Governed Space: Simulation Parameterization

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AI-Readable: Yes

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Canonical Language: English (UCL™)

Governed Space

Simulation Parameterization

Canonical Definition

Simulation Parameterization Module (SPHM) defines the governance framework for identifying, defining, assigning, constraining, documenting, controlling, changing, and reconstructing the parameters through which a governed Simulation Model and Simulation Environment are configured for simulation execution.

SPHM establishes the controlled parameter layer between the structural design of the simulation and the specific configured conditions under which simulation behavior is produced.

Operational Role

SPHM is the fourth internal governance space of the Simulation Model & Environment Standard (SMES).

The preceding modules establish:

SMMM — What is the Simulation Model structurally composed of?

SECM — Within what configured Simulation Environment does that model operate?

SBIM — How do the model, environment, and connected domains interact across their boundaries?

SPHM asks:

Which configurable values govern the behavior of the model and environment, why were those values selected, what ranges and constraints apply, and which parameter configuration produced a particular simulation execution?

The governed progression is:

Model Structure → Environment Configuration → Boundary Interaction → Parameter Identification → Parameter Definition → Value Assignment → Constraint & Range Control → Parameter Configuration → Execution Binding → Governed Simulation Parameterization

Module Operational Space

SPHM governs: Simulation Parameter identification, parameter identity, parameter definition, parameter classification, parameter ownership, parameter source, parameter basis, parameter units, parameter types, parameter values, parameter ranges, parameter distributions, parameter constraints, parameter dependencies, parameter relationships, parameter inheritance, parameter overrides, default parameters, fixed parameters, variable parameters, dynamic parameters, stochastic parameters, derived parameters, calibrated parameters, externally supplied parameters, model parameters, environment parameters, boundary parameters, parameter sets, parameter configurations, parameter baselines, parameter sensitivity, parameter change, parameter substitution, parameter uncertainty references, parameter traceability, execution binding, and parameter reconstruction.

Module Operational Scope

SPHM applies wherever configurable values materially influence simulation behavior.

This may include: computational simulations, engineering simulations, robotics simulations, autonomous-system simulations, AI simulations, digital twins, scientific models, climate and environmental simulations, industrial simulations, infrastructure simulations, financial simulations, economic simulations, biological simulations, healthcare simulations, transportation simulations, aerospace simulations, mission simulations, multi-agent simulations, software-in-the-loop environments, hardware-in-the-loop environments, and hybrid physical-digital simulations.

Governance depth should remain proportional to the material influence of parameterization upon the simulation inquiry.

Module Function

A correctly structured model can produce materially different behavior depending upon how it is parameterized.

For example:

Same Model + Different Parameters = Different Simulation Behavior

A simulation may therefore fail methodologically even when its model structure is correct if: material parameters are undocumented, default values are silently inherited, parameter values lack an identifiable basis, permissible ranges are undefined, incompatible parameters are combined, units are inconsistent, values change during execution without visibility, parameter dependencies are ignored, or the parameter set used for a result cannot later be reconstructed.

SPHM prevents parameterization from becoming an invisible technical configuration layer.

Simulation Parameter

A Simulation Parameter is a governed configurable property whose assigned value, range, distribution, rule, or state may materially influence the behavior of a Simulation Model, Simulation Environment, Simulation Boundary, or simulation execution.

A parameter may represent: physical property, behavioral property, environmental condition, system capability, timing property, probability, threshold, tolerance, resource quantity, interaction property, control value, or another configurable simulation characteristic.

Parameter Identity

Each material parameter should be sufficiently identifiable.

Where relevant, Parameter Identity may establish: parameter name, parameter identifier, parameter type, governed component, unit, source, basis, permissible range, current value, and applicable configuration.

Parameter names alone may be insufficient where multiple components use similarly named variables.

Parameter Definition

A parameter should have a sufficiently precise definition to establish what the value actually represents.

For example:

Reaction Time = 0.8

is incomplete if the simulation record does not establish: whose reaction time, measured in which unit, under which conditions, applied to which model component, and whether the value is fixed, variable, or distributed.

Parameter Classification

Parameters may be classified according to their simulation role.

Possible classifications include:

Model Parameter

Environment Parameter

Boundary Parameter

Control Parameter

Initial-State Parameter

Behavioral Parameter

Physical Parameter

Timing Parameter

Resource Parameter

Stochastic Parameter

or another simulation-specific class.

Model Parameter

A Model Parameter configures a property of the Simulation Model without changing its underlying structural identity unless the change becomes structurally material.

Examples may include: mass, acceleration limit, control threshold, sensor range, decision threshold, response rate, or behavioral coefficient.

Environment Parameter

An Environment Parameter configures a governed property of the Simulation Environment.

Examples may include: temperature, traffic density, visibility, network latency, market liquidity, resource availability, or environmental response rate.

SECM establishes the environment and its configuration structure.

SPHM governs the controlled values through which configurable environmental properties are instantiated.

Boundary Parameter

A Boundary Parameter configures a property of a governed Simulation Boundary or interaction mechanism.

Examples may include: interface latency, transmission frequency, timeout, conversion tolerance, synchronization interval, or signal range.

SBIM governs the boundary interaction.

SPHM governs its configurable parameter values.

Parameter Type

A parameter may be:

Fixed

Variable

Dynamic

Stochastic

Derived

Calibrated

Externally Supplied

Scenario-Controlled

or:

Execution-Controlled

where the distinction materially affects simulation behavior.

Fixed Parameter

A Fixed Parameter is intended to remain unchanged during the applicable simulation execution or governed simulation configuration.

Its fixed status should not imply that its value requires no basis.

Variable Parameter

A Variable Parameter may take different governed values between simulation runs or configurations.

The permissible variation should remain bounded where material.

Dynamic Parameter

A Dynamic Parameter may change during simulation execution according to defined logic, state, events, or external conditions.

Its governing rule may be as important as its initial value.

Stochastic Parameter

A Stochastic Parameter is governed through a probability distribution or stochastic mechanism rather than one fixed value.

Relevant governance may include: distribution type, parameters of the distribution, permitted range, sampling rule, correlation, and seed relationship where material.

Derived Parameter

A Derived Parameter is calculated from other governed values.

Conceptually:

Parameter A + Parameter B + Derivation Rule → Derived Parameter C

The derivation should remain traceable where the resulting parameter materially affects simulation behavior.

Calibrated Parameter

A Calibrated Parameter receives its value through a calibration process.

SPHM governs the parameterization role and resulting value.

It does not replace broader validation governance over whether the calibration evidence is sufficient for an intended validation claim.

Externally Supplied Parameter

Some parameters originate outside the simulation.

Examples include: operational measurements, external datasets, sensor values, market values, environmental feeds, engineering specifications, or third-party model outputs.

Their external origin should remain identifiable.

Parameter Basis

Every material parameter should have an identifiable basis where practicable.

A Parameter Basis may derive from: measured reality, scientific literature, engineering specification, historical data, expert judgment, Simulation Assumption, calibration, scenario design, operational data, regulatory requirement, or explicit experimental selection.

Parameter Basis ≠ Parameter Validity

SPHM governs:

Why was this parameter value selected and how is it represented inside the

simulation?

VALIDOS® may separately govern:

Is the evidentiary and methodological basis sufficient to validate reliance upon the resulting simulation evidence?

This separation protects the architectural boundary between SIMULOS® and VALIDOS®.

Assumption-to-Parameter Traceability

Simulation Assumptions may become parameter values.

Conceptually:

Simulation Assumption → Parameter Definition → Parameter Value

For example:

Assumption: Average Human Reaction Time = 1 second

may become:

Reaction_Time = 1.0 s

Where material, the relationship should remain traceable.

Reality-Representation-to-Parameter Traceability

Reality Representation requirements may also determine parameterization.

Conceptually:

Represented Reality Property → Simulation Variable → Parameter Definition → Parameter Value / Distribution

This establishes continuity between what the simulation claims to represent and how that representation is configured.

Parameter Value

A Parameter Value is the assigned value or state applied to a parameter within a defined Simulation Configuration.

A value should remain interpretable together with: parameter identity, unit, applicable range, source or basis, configuration, and execution context.

Parameter Units

Where parameters represent measurable quantities, units should be explicit.

Examples:

Velocity = 30 m/s

Latency = 120 ms

Temperature = 25 °C

Mass = 1,500 kg

Unit ambiguity can materially alter simulation behavior.

Unit Compatibility

Parameters participating in shared calculations or interactions should use compatible units or governed conversion.

Silent unit conversion should not create an invisible simulation transformation.

Parameter Range

A Parameter Range establishes the values within which a parameter may legitimately operate for a defined simulation configuration.

Conceptually:

Minimum ≤ Parameter ≤ Maximum

Ranges may be: physical, methodological, empirical, scenario-defined, operational, safety-related, or technically imposed.

Parameter Range Violation

A parameter value outside its governed range should not silently enter simulation execution.

Possible treatments may include:

Reject

Flag

Constrain

Require Authorization

Create Exception

or another governed response.

Parameter Constraint

A Parameter Constraint defines a condition restricting permissible parameter values or combinations.

Constraints may include: minimum, maximum, allowed states, prohibited combinations, conditional ranges, dependency conditions, or cross-parameter relationships.

Cross-Parameter Constraint

Some values may be legitimate individually but illegitimate together.

For example:

Payload Mass = Valid

and:

Fuel Level = Valid

may nevertheless create an impossible combined configuration.

SPHM therefore governs parameter relationships as well as individual values.

Parameter Dependency

A Parameter Dependency exists where the value, range, interpretation, or effect of one parameter depends materially upon another parameter or governed simulation condition.

Conceptually:

Parameter A → constrains / determines / modifies → Parameter B
Parameter Dependency Map

Where dependencies are material, SPHM should support:

Parameter → Dependency → Dependent

Parameter → Effect

This becomes particularly important in complex models containing hundreds or thousands of configurable variables.

Parameter Distribution

Some simulation properties are more appropriately represented by distributions than single values.

Examples include: human reaction times, equipment failure rates, weather variation, demand behavior, network latency, biological variability, or market behavior.

The distribution itself becomes a governed parameterization object.

Distribution Governance

Where material, governance may establish: distribution type, central tendency, dispersion, bounds, truncation, correlation, sampling method, and applicable population or condition.

Parameter Correlation

Parameters may be statistically or structurally related.

Treating correlated parameters as independent may create unrealistic simulation states.

Material correlation assumptions should therefore remain visible.

Default Parameter

A Default Parameter is a parameter value automatically applied when no alternative value is explicitly selected.

Defaults are not methodologically neutral.

A default can materially shape simulation behavior.

No Invisible Default Principle

A material parameter must not escape governance merely because its value was supplied automatically by software, inherited from a template, or left unchanged from a previous configuration.

Parameter Inheritance

Parameters may be inherited from: parent configurations, templates, model defaults, environment defaults, prior simulation versions, or scenario configurations.

Material inheritance should remain traceable.

Parameter Override

A Parameter Override occurs when an inherited, default, or previously established value is replaced for a specific configuration or execution.

An override should identify: affected parameter, previous value, new value, reason, applicable scope, and execution relationship

where material.

Parameter Set

A Parameter Set is a governed collection of parameters intended to operate together.

Examples:

Baseline Parameter Set

Adverse Weather Parameter Set

High-Load Parameter Set

Failure Parameter Set

Experimental Parameter Set Parameter Configuration

A Parameter Configuration is the complete materially relevant combination of parameter values, distributions, constraints, dependencies, and overrides applicable to a defined simulation execution or group of executions.

Conceptually:

Simulation Parameter Configuration = Parameter Set + Values + Ranges + Constraints + Dependencies + Overrides Parameter Configuration Identifier

Material parameter configurations should be uniquely distinguishable where different configurations may produce different simulation behavior.

This enables:

Simulation Result → Simulation Execution → Parameter Configuration Parameter Baseline

A Parameter Baseline establishes the approved reference parameter configuration against which changes may be identified.

The baseline may support: repeatability, comparability, sensitivity analysis, scenario variation, configuration control, and change detection.

Baseline ≠ Universal Correctness

A baseline establishes a controlled reference.

It does not establish that the baseline perfectly represents reality.

Parameter Variation

Parameter Variation may be intentional.

Examples include: sensitivity testing, stress testing, scenario exploration, uncertainty exploration, optimization, robustness testing, or boundary testing.

SPHM governs the variation itself.

Scenario Design governance determines how parameter variation is organized into governed scenarios.

Parameter Variation Range

Where parameter variation is intentional, the tested range should remain identifiable.

For example:

Wind Speed: 0–150 km/h

rather than simply:

Wind varied.

Parameter Sweep

A Parameter Sweep systematically varies one or more parameters across defined values or ranges.

SPHM may govern: parameter identity, variation range, step, distribution, combinations, exclusions, and resulting configuration identifiers.

Parameter Sensitivity

Parameter Sensitivity describes the degree to which changes in a parameter materially alter simulation behavior or outputs.

SPHM governs the identification and configuration relevance of sensitivity.

It does not replace the downstream interpretation or validation of simulation outputs.

High-Sensitivity Parameter

A parameter may be classified as high-sensitivity where small changes can materially alter simulation behavior.

Such parameters may justify stronger: basis documentation, range control, change control, traceability, and execution binding.

Parameter Sensitivity ≠ Uncertainty Governance

Sensitivity asks:

How strongly does simulation behavior respond when this parameter changes?

Uncertainty asks:

How uncertain are we about the parameter or its basis?

Simulation Assumption governance and other relevant spaces may govern uncertainty.

SPHM preserves the parameterization relationship without duplicating those

spaces.

Parameter Substitution

A parameter may be substituted when the originally intended value or source is unavailable.

For example:

Measured Wind Data → Synthetic Wind Distribution

The substitution should remain visible where material.

Parameter Change

A Parameter Change may involve: value change, range change, distribution change, unit change, constraint change, dependency change, source change, basis change, inheritance change, override, or parameter removal.

Parameter Change Record

Material changes should preserve:

Previous Parameter State → Change → New Parameter State → Applicable Configuration Parameter Drift

Parameter Drift occurs where parameter values or parameterization behavior progressively depart from the governed configuration without an equivalent governed decision.

Drift may arise through: automated adaptation, external feeds, configuration changes, software updates, calibration updates, environment changes, or inherited defaults.

Dynamic Parameter Change

A parameter changing during execution according to its governed rule is not necessarily drift.

The distinction is:

Governed Dynamic Change

versus:

Uncontrolled Parameter Change Parameter Lock

A parameter may be locked where its value must not change during a defined execution or comparison set.

Locking may be important for: repeatability, controlled experimentation,

comparative simulation, benchmark execution, or forensic reconstruction.

Parameter Freeze

A Parameter Freeze may preserve a complete parameter configuration for a defined simulation execution campaign.

This creates a stable configuration reference.

Parameterization and Scenario Design

SPHM governs:

Which parameters exist and how their values may be configured.

Scenario Design governs:

Which combinations of conditions and parameter configurations are intentionally selected as scenarios for simulation.

Therefore:

Parameterization ≠ Scenario Design

This boundary prevents duplication between Parent Standards.

Parameterization and Input Data

A parameter should also remain distinguishable from Input Data.

A parameter configures the simulation.

Input Data supplies data consumed during simulation execution.

In some systems the distinction may be technically blurred.

The governance classification should follow operational function rather than software implementation.

Parameterization and Assumptions

Simulation Assumptions may justify or constrain parameter values.

However:

Assumption Governance ≠ Parameter Governance

SAS governs the assumption and its role.

SPHM governs how that assumption becomes an operational simulation configuration.

Parameterization and Environment Configuration

SECM establishes:

what environmental configuration exists.

SPHM establishes:

which configurable values instantiate or control material properties of that environment.

Parameterization and Boundary Interaction

SBIM establishes:

what may cross a Simulation Boundary and under which interaction conditions.

SPHM governs:

the configurable values controlling those interaction properties.

Parameter Configuration Completeness

Parameter Configuration Completeness asks:

Have all material parameters required to reconstruct the governed simulation execution been identified and preserved?

Completeness does not require documenting every technically available setting.

It requires visibility over settings capable of materially affecting the simulation inquiry.

Parameter Configuration Consistency

Parameter Configuration Consistency asks whether: units are compatible, values fall within applicable ranges, dependencies are satisfied, constraints are respected, inherited values remain applicable, and combined values form a coherent simulation configuration.

Parameter Configuration Coherence

A parameter configuration may contain individually valid values while producing an incoherent overall simulation state.

SPHM therefore evaluates configuration-level coherence rather than only individual parameter validity.

Parameter Comparability

Two simulation executions should not be treated as parameter-equivalent unless their materially relevant parameter configurations are sufficiently aligned.

Differences should remain visible where they affect interpretation.

Controlled Comparison

For controlled comparative simulation:

Compared Executions should differ only in the variables intentionally changed or otherwise explicitly disclosed.

This enables parameter-level causal interpretation without assuming broader methodological validation.

Parameter Change Propagation

A material parameter change may propagate through later SIMULOS® spaces.

Conceptually:

Parameter Change

↓

Scenario Configuration

↓

Simulation Execution

↓

Simulation Output

↓

Reality Correlation

↓

Simulation Evidence

SPHM preserves the parameter change.

The downstream spaces govern their respective consequences.

Parameter-to-Execution Binding

Every material simulation execution should be bindable to the Parameter Configuration under which it occurred.

Conceptually:

Simulation Execution ID → Parameter Configuration ID

Without this relationship, repeatability and reconstruction may become impossible.

Parameter-to-Output Traceability

Where material, simulation outputs should remain traceable to the parameter configuration that contributed to their production.

Conceptually:

Parameter Configuration → Execution → Simulation Output

SPHM establishes the parameter side of that traceability chain.

Parameter Reconstruction

SPHM should enable reconstruction sufficient to answer:

Which parameters were active?

Which values were assigned?

Which units applied?

Which ranges and constraints applied?

Which values were inherited or overridden?

Which parameters were dynamic or stochastic?

Which dependencies existed?

Which parameter configuration produced the execution?

Did any material parameter change during execution?

Parameterization Traceability

A complete parameterization traceability path may be represented as:

Simulation Purpose & Scope

↓

Reality Representation

↓

Simulation Assumptions



Simulation Model & Environment



Simulation Parameter



Parameter Basis



Parameter Value / Distribution



Parameter Configuration



Simulation Execution

Minimum Implementation Framework

1. Identify Material Simulation Parameters

Determine which configurable properties may materially influence simulation behavior. 2. Define and Classify Parameters

Establish meaning, type, unit, role, and governed component. 3. Establish Parameter Basis

Identify the basis supporting material parameter values or distributions. 4. Establish Values, Ranges, and Distributions

Define applicable values and permissible variation. 5. Govern Constraints and Dependencies

Identify relationships that limit or modify parameter configuration. 6. Control Defaults, Inheritance, and Overrides

Prevent invisible parameterization through technical defaults or inherited configurations. 7. Establish Parameter Sets and Baselines

Create controlled parameter configurations for simulation activity. 8. Govern Material Parameter Change

Preserve changes, substitutions, drift, and dynamic behavior. 9. Bind Parameter Configuration to Execution

Ensure each material execution can identify its applicable parameter configuration. 10. Preserve Parameter Reconstruction

Maintain sufficient traceability to reconstruct the parameterization that produced simulation behavior.

Governance Outputs

SPHM may produce: Simulation Parameter Register, Simulation Parameter Identifier, Parameter Definition Record, Parameter Classification Record, Parameter Basis Record, Parameter Source Record, Parameter Unit Record, Parameter Value Record, Parameter Range Record, Parameter Distribution Record, Parameter Constraint Record, Cross-Parameter Constraint Record, Parameter Dependency Map, Parameter Correlation Record, Default Parameter Record, Parameter Inheritance Record, Parameter Override Record, Parameter Set, Parameter Configuration, Parameter Configuration Identifier, Parameter Baseline, Parameter Variation Record, Parameter Sweep Record, Parameter Sensitivity Record, High-Sensitivity Parameter Record, Parameter Substitution Record, Parameter Change Record, Parameter Drift Record, Parameter Lock Record, Parameter Freeze Record, Assumption-to-Parameter Traceability Map, Reality-Representation-to-Parameter Traceability Map, Parameter Configuration Completeness Assessment, Parameter Configuration Consistency Assessment, Parameter Configuration Coherence Assessment, Parameter-to-Execution Binding, Parameter Reconstruction Record, and Simulation Parameterization Traceability Map.

Use Case 1 — Autonomous Vehicle Simulation

Scenario

An autonomous vehicle simulation uses configurable values for:

Vehicle Mass + Sensor Range + Reaction Delay + Road Friction + Visibility + Traffic Density Application

SPHM identifies each material parameter, establishes units and ranges, preserves the source or assumption supporting each value, and binds the resulting Parameter Configuration to the simulation execution.

A change from:

Road Friction = 0.8

to:

Road Friction = 0.3

creates a materially different parameter configuration. Result

The resulting vehicle behavior can be traced to the exact parameter configuration under which it was produced. Use Case 2 — Spacecraft Mission Simulation

Scenario

A spacecraft simulation includes:

Vehicle Mass + Fuel Load + Engine Performance + Communication Delay + Thermal Conditions + Orbital Parameters Application

SPHM governs the values, units, permissible ranges, dependencies, and parameter sets.

If spacecraft mass changes, dependent fuel and propulsion parameters can be identified rather than remaining silently inconsistent. Result

Mission simulation executions remain reconstructable and parameter-comparable.
Use Case 3 — Financial Stress Simulation

Scenario

A financial simulation varies:

Interest Rate + Default Probability + Liquidity + Asset Correlation + Market Volatility Application

SPHM governs parameter ranges, distributions, correlations, parameter sweeps, and the specific configuration applied to each execution. Result

A stress result can be distinguished from a baseline result through its governed Parameter Configuration rather than only through its final output.

Architectural Position

Simulation Parameterization is the fourth internal governance space of the Simulation Model & Environment Standard (SMES).

The complete SMES progression is:

Simulation Model Structure → Simulation Environment Configuration → Simulation Boundary Interaction → Simulation Parameterization → Simulation Environment Versioning

SPHM receives: the governed Simulation Model Structure from SMMM, the governed Simulation Environment Configuration from SECM, and the governed Simulation Boundary Interaction structure from SBIM.

It then establishes the controlled parameter configuration required to determine which specific configuration of the model and environment is actually prepared for simulation activity.

The resulting governed configuration becomes an input to Simulation Environment Versioning and later SIMULOS® lifecycle spaces.

Foundational Principle

A simulation cannot be meaningfully reconstructed, compared, or governed when the material parameters shaping its behavior are unknown, uncontrolled, or detached from the execution they produced.

Canonical Closing Statement

Simulation Parameterization Module establishes the parameter governance layer of the Simulation Model & Environment Standard by transforming configurable model, environment, and boundary properties into explicit, bounded, traceable, and reconstructable Simulation Parameter Configurations. Through parameter identification, definition, classification, basis, units, values, ranges, distributions, constraints, dependencies, defaults, inheritance, overrides, sensitivity, controlled variation, change governance, configuration baselines, and execution binding, SPHM ensures that materially different simulation behavior cannot be produced by invisible parameter choices while being represented as if it arose from the same governed simulation configuration.

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